

Machine Learning: Regularization & Variable Selection

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Regularization

Regularization

Problem of overfitting: Noise in the data is spuriously fit as systematic pattern.

Regularization solution: Modify the estimation criterion to penalize explaining variation as systematic.

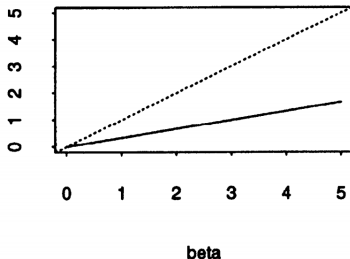
Canonical approaches: Ridge, lasso, and elasticnet methods for regression.

Many extensions/applications: e.g., gene network inference, clustering, topic models, factor analysis.

Regularization: Ridge regression

Ridge: “L2” penalization of coefficient magnitudes.

$$\hat{\beta} = \min_{\beta} \left[\sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right]$$

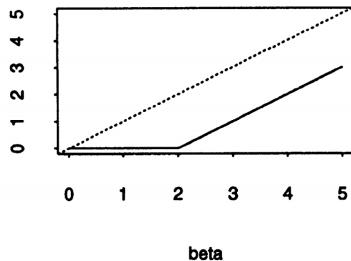


Source: Tibshirani, R., 1996. Regression shrinkage and selection via the lasso. Journal of the Royal Statistical Society: Series B (Methodological), 58(1), pp.267-288. Vancouver

Regularization: Lasso regression

Lasso: “L1” penalization of coefficient magnitudes.

$$\hat{\beta} = \min_{\beta} \left[\sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2 + \lambda \sum_{j=1}^p |\beta_j| \right]$$



Source: Tibshirani, R., 1996. Regression shrinkage and selection via the lasso. Journal of the Royal Statistical Society: Series B (Methodological), 58(1), pp.267-288. Vancouver

Regularization: Elasticnet regression

Elasticnet: “L1” and “L2” penalization combined.

$$\hat{\beta} = \min_{\beta} \left[\sum_{i=1}^n (y_i - \mathbf{x}_i' \beta)^2 + \lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=1}^p \beta_j^2 \right]$$

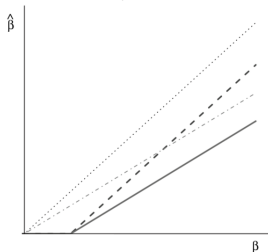


Fig. 2. Exact solutions for the lasso (-----), ridge regression (·-·-·-·) and the naïve elastic net (——) in an orthogonal design (·····, OLS): the shrinkage parameters are $\lambda_1 = 2$ and $\lambda_2 = 1$

Source: Zou, H. and Hastie, T., 2005. Regularization and variable selection via the elastic net. Journal of the royal statistical society: series B (statistical methodology), 67(2), pp.301-320.. Vancouver

Considerations

Tuning and selection

- ▶ Cross-validation is most common approach for tuning λ 's.
- ▶ Elasticnet usually best, but takes much longer to tune.

Variable selection

- ▶ Lasso's strength is the boundary solution (i.e., $\hat{\beta} = 0$).
- ▶ Challenge: Regularization parameters optimized for prediction.
- ▶ Key property: One-shot lasso over-selects variables relative to the true model.
Two solutions:
 - ▶ Double lasso: Belloni, A., Chernozhukov, V., Hansen, C., 2014. Inference on treatment effects after selection among high-dimensional controls. The Review of Economic Studies.
 - ▶ Bootstrap lasso: Bach, F.R., 2008, July. Bolasso: model consistent lasso estimation through the bootstrap. ICML 25.

Into RStudio

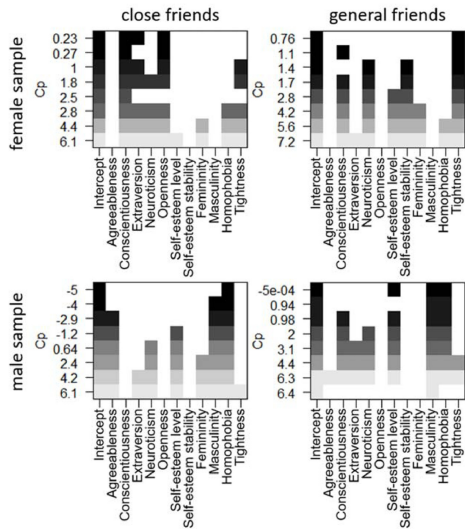
Best Subsets Regression

Overview of Best Subsets Regression

- ▶ Examines all possible combinations of predictors to identify the optimal model.
- ▶ Aims to find a balance between model complexity and explanatory power.
- ▶ Valuable when we have numerous potential predictors.
- ▶ Fewer predictors can aid interpretation.

Why Best Subsets Regression?

- ▶ Unlike, e.g., stepwise regression or LASSO, best subsets regression evaluates every combination of predictors.
- ▶ Ensures we don't miss the best model.
- ▶ Conceptually ideal approach to both variable and model selection.



Altmann, T. and Roth, M., 2020. Individual differences in the preference for cross-sex friendship (heterosociality) in relation to personality. *Personality and Individual Differences*, 157, p.109838.

Algorithm of Best Subsets Regression

The algorithm involves:

1. Enumerating all possible predictor combinations.
2. For each combination, fitting a model and calculating its performance using a chosen metric.
3. Comparing models to select the one with the best performance.

This process, though computationally demanding, ensures a comprehensive evaluation of the model space.

Computational Complexity

The computational intensity of the brute-force approach to best subsets regression increases exponentially with the number of predictors.

- ▶ Represented in Big O notation, $O(2^n)$, where n is the number of predictors.
- ▶ This exponential growth poses significant challenges when dealing with large datasets.

Fast Best Subsets

PNAS | NOW READING: **A polynomial algorithm for best-subset selection problem**

RESEARCH ARTICLE | PHYSICAL SCIENCES |

A polynomial algorithm for best-subset selection problem

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Edited by Runze Li, Pennsylvania State University, State College, PA, and accepted by Editorial Board Member David A. Weitz November 18, 2020 (received for review July 7, 2020)

December 16, 2020 | 117 (52) 33117-33123 | <https://doi.org/10.1073/pnas.2014241117>

12,127 | 17



Significance

Best-subset selection is a benchmark optimization problem in statistics and machine learning. Although many optimization strategies and algorithms have been proposed to solve this problem, our splicing algorithm, under reasonable conditions, enjoys the following properties simultaneously with high probability: 1) its computational complexity is polynomial; 2) it can recover the true subset; and 3) its solution is globally optimal.



ABESS overview

Let $\|\beta\|_0 = \sum_{i=1}^p I(\beta_p \neq 0)$. Consider the problem

$$\min_{\beta} L_n(\beta) \quad \text{s.t.} \|\beta\|_0 \leq s$$

, where $L_n(\beta)$ is $1/(2n)$ SSE

1. At a given s , use some screening method to define the active set as the variables for which $\beta_p \neq 0$, and the inactive set as the variables for which $\beta_p = 0$.
2. For values of $k \in (1, 2, \dots, s)$, splice the best k inactive variables with the worst k active variables, until unable to improve by τ .
3. Repeat 1 & 2 for all values of $s \leq \frac{n}{\ln(p) \ln[\ln(n)]}$
4. Select subset w.r.t. “special” information criterion

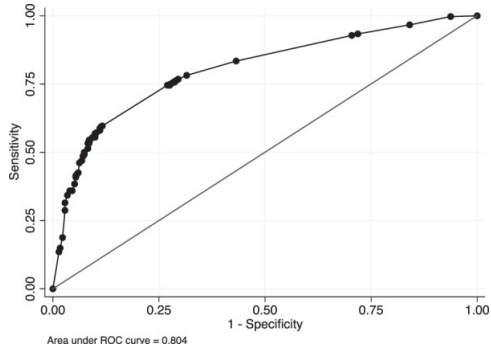
$$\text{SIC} = nL_n(\beta) + \|\beta\|_0 \ln(p) \ln [\ln(n)]$$

Table 3*Bivariable and Multivariable Linear Regression Analyses Examining Factors Associated With Depressive Symptoms*

Explanatory variable	Bivariable Analysis			Multivariable Analysis		
	β	<i>SE</i>	<i>p</i>	β	<i>SE</i>	<i>p</i>
Psychological inflexibility	.406	.028	<.001***	.378	.028	<.001***
Employment status	2.84	1.19	.019*	-1.39	.650	.035*
Time since the injury	-.188	.044	<.001***	-.055	.027	.047*
Age at the time of the injury	.132	.043	.003**			
Number of symptoms related to SCI	.857	.474	.074			
Pain interference	.222	.039	<.001***			
Independence	-.295	.054	<.001***			
Satisfaction with social roles and activities	-.637	.097	<.001***			

Note. SCI = spinal cord injury; *SE* = standard error.**p* < .05. ***p* < .001. ****p* < .001.

Han, A., Yuen, H.K. and Wilroy, J.D., 2022. Exploring associations of psychological inflexibility with depressive symptoms, anxiety, and stress in individuals living with a spinal cord injury: A cross-sectional study. *Rehabilitation psychology*, 67(2), p.170.



Koehler, T.J., Goodfellow, J., Davis, A.T., Spybrook, J., vanSchagen, J.E. and Schuh, L., 2017. Predicting in-state workforce retention after graduate medical education training. *Journal of Graduate Medical Education*, 9(1), pp.73-78.

Variable	β Coefficient	Score Computation	Rounded Score
Born in Michigan	0.889		
Yes = 1	$1 * 0.889 = 0.889$	$0.889/0.593 = 1.499$	1.5
No = 0 ^a	$0 * 0.889 = 0$	$0/0.593 = 0$	0
Primary care program graduate	0.768		
Yes = 1	$1 * 0.768 = 0.768$	$0.768/0.593 = 1.295$	1.5
No = 0 ^a	$0 * 0.768 = 0$	$0/0.593 = 0$	0
Completed undergraduate education in Michigan	1.135		
Yes = 1	$1 * 1.135 = 1.135$	$1.135/0.593 = 1.914$	2
No = 0 ^a	$0 * 1.135 = 0$	$0/0.593 = 0$	0
Ever married	0.698		
Yes = 1	$1 * 0.698 = 0.698$	$0.698/0.593 = 1.177$	1
No = 0 ^a	$0 * 0.698 = 0$	$0/0.593 = 0$	0
Completed medical school in Michigan	0.901		
Yes = 1	$1 * 0.901 = 0.901$	$0.901/0.593 = 1.519$	1.5
No = 0 ^a	$0 * 0.901 = 0$	$0/0.593 = 0$	0
Completed GME training in Michigan	0.593		
Yes = 1	$1 * 0.593 = 0.593$	$0.593/0.593 = 1.000$	1
No = 0 ^a	$0 * 0.593 = 0$	$0/0.593 = 0$	0

Abbreviation: GME, graduate medical education.

^a Denotes reference category.

Koehler, T.J., Goodfellow, J., Davis, A.T., Spybrook, J., vanSchagen, J.E. and Schuh, L., 2017. Predicting in-state workforce retention after graduate medical education training. *Journal of Graduate Medical Education*, 9(1), pp.73-78.

Takeaways

- ▶ Regression is a powerful and foundational tool in machine learning.
- ▶ Regularization optimizes regression estimates for predictive performance.
- ▶ Polynomial regression can be used to find flexible regression functions.
- ▶ GAMs offer a state-of-the-art approach to using polynomials for nonlinear model-fitting.
- ▶ Best subsets regression represents an effective approach to variable selection.

Key readings

- ▶ Dalal, Siddhartha R., Edward B. Fowlkes, and Bruce Hoadley. (1989). Risk analysis of the space shuttle: Pre-Challenger prediction of failure. *Journal of the American Statistical Association*, 84(408), 945-957.
- ▶ Michener, Jamila. (2013). Neighborhood disorder and local participation: Examining the political relevance of “broken windows”. *Political Behavior*, 35, 777-806.
- ▶ Hastie, Trevor, and Robert Tibshirani. (1987). Generalized additive models: some applications. *Journal of the American Statistical Association*, 82(398), 371-386.
- ▶ Zhu, Jin, Xueqin Wang, Liyuan Hu, Junhao Huang, Kangkang Jiang, Yanhang Zhang, Shiyun Lin, and Junxian Zhu. "abess: A fast best-subset selection library in python and r." *Journal of Machine Learning Research* 23, no. 202 (2022): 1-7.
- ▶ Tibshirani, Robert. "Regression shrinkage and selection via the lasso." *Journal of the Royal Statistical Society Series B: Statistical Methodology* 58, no. 1 (1996): 267-288.

If we have time:
Applications

How to read these application spotlights

- ▶ Each example starts from a **published study whose main analysis is a regression**.
- ▶ We **reproduce** the original fit, then ask a prediction question and **extend with an ML learner**.
- ▶ All datasets come from public replication archives (Dataverse / OSF), so every result here is reproducible.
- ▶ Takeaway each time: regression answers the paper's question; the learner often *predicts* better.

Application 1

Predicting state legislative-election turnout

Kuhlmann & Lewis (2017), *State Politics & Policy Quarterly*

The study and the data

- ▶ **Question:** do legislative term limits depress voter turnout?
- ▶ **Original analysis:** OLS of state lower-chamber turnout on 13 predictors (term-limitedness, professionalism, party competition, registration rules, diversity, income, year, ...).
- ▶ **Data:** UNC Dataverse, doi:10.15139/S3/POGNLC ($n = 486$ state-years).
- ▶ We reproduce the OLS, then compare its *out-of-sample* accuracy to Elastic Net and XGBoost.

Reproducing the published model

- ▶ OLS reproduces cleanly: **in-sample** $R^2 = 0.723$ (adj. $R^2 = 0.715$), $n = 486$.
- ▶ The coefficients recover the paper's substantive story — this is the right model for the *explanatory* question.
- ▶ But “fits in sample” $\neq$ “predicts out of sample.” We test honestly with 10-fold cross-validation.

Out-of-sample: OLS vs. regularization vs. boosting

10-fold cross-validated prediction of turnout (%)

Model	CV RMSE	CV R^2
OLS	5.69	0.71
Elastic Net	5.68	0.71
XGBoost	4.38	0.83

- ▶ XGBoost cuts out-of-sample RMSE by **23%** over OLS.
- ▶ Elastic Net $\approx$ OLS here: shrinkage alone does not help when p is small and signal is dense; the gain comes from **nonlinearity & interactions** the trees capture.

Takeaway — turnout

- ▶ Same data, two jobs: OLS for *interpretation*, boosting for *prediction*.
- ▶ Reporting both is honest and informative: “the linear model explains X; a flexible learner predicts Y% better out of sample, suggesting unmodeled nonlinearity.”
- ▶ Code: `tutorials/part2/linear_model/regression_vs_ml_turnout.R`.

Application 2

Variable selection in personality research

Altmann & Roth (2020), *Personality and Individual Differences*

From a regression table to a selection problem

- ▶ **Question:** which personality facets predict the preference for cross-sex friendship (heterosociality)?
- ▶ **Original analysis:** multiple regression with a large facet battery — a natural **variable-selection** setting.
- ▶ Many correlated predictors $\Rightarrow$ unstable coefficients and over-fitting risk.

Extending with principled selection

- ▶ **Lasso / Elastic Net:** shrink and drop facets; tune the penalty by cross-validation.
- ▶ **Best subsets / ABESS:** search the model space for the most predictive small set.
- ▶ **Random-forest importance:** a nonparametric check on which facets carry predictive weight.
- ▶ Compare the selected sets — agreement across methods is reassuring; disagreement is informative.

Takeaway — selection

- ▶ Stepwise p -value selection is fragile; lasso/ABESS optimize selection for *prediction*.
- ▶ Caveat already covered: one-shot lasso *over-selects* — use double or bootstrap lasso for inference.
- ▶ Good teaching contrast with the in-class abess demonstration.

Application 3

Predicting workforce retention

Koehler et al. (2017), *J. of Graduate Medical Education*

A management-flavored prediction task

- ▶ **Question:** which trainees stay and practice in-state after graduate medical education?
- ▶ **Original analysis:** logistic regression / best-subsets on program + trainee characteristics.
- ▶ Binary outcome, modest n , many candidate predictors — a recurring shape in HR / workforce analytics.

Extending with ML + honest evaluation

- ▶ Penalized logistic (lasso) for a parsimonious, regularized model.
- ▶ Random forest / gradient boosting for nonlinear interactions among predictors.
- ▶ Because retention is imbalanced, evaluate with **ROC-AUC** and **PR-AUC**, not accuracy.
- ▶ Report which predictors the selection methods retain — actionable for program design.

Takeaway — retention

- ▶ The same regularization + selection toolkit transfers directly from political/health data to HR and management questions.
- ▶ Lesson: choose the metric to match the (imbalanced) decision problem before comparing models.

Application 4

Psychological inflexibility and distress

Han, Yuen & Wilroy (2022), *Rehabilitation Psychology*

Several outcomes, shared predictors

- ▶ **Question:** how does psychological inflexibility relate to depression, anxiety, and stress among people with spinal-cord injury?
- ▶ **Original analysis:** multiple regression, one per outcome.
- ▶ Cross-sectional, correlated predictors — prime ground for regularization.

Extending the analysis

- ▶ Ridge/Elastic Net stabilizes estimates when predictors are collinear.
- ▶ A GAM allows the inflexibility→distress relationship to bend, rather than forcing a line.
- ▶ Cross-validate each outcome; report the predictive gain (if any) honestly — sometimes the linear model is already adequate, and that is a finding too.

Section recap — when does ML help?

- ▶ **Big gains** when the truth is nonlinear / interactive and n is reasonable (e.g. turnout: -23% RMSE).
- ▶ **Small gains** when the linear model is already close (dense, near-linear signal, small p).
- ▶ Regularization buys stability and selection; trees/boosting buy flexibility.
- ▶ Always decide with **out-of-sample** evidence, and match the metric to the problem.

Application 5

Predicting civil-war onset — the canonical case

Fearon & Laitin (2003), *APSR*; cf. Muchlinski et al. (2016), *Political Analysis*

The study and the data

- ▶ **Original analysis:** logistic regression of civil-war *onset* on ~ 15 country-year covariates (income, population, terrain, instability, ...).
- ▶ **Data:** public replication archive on Harvard Dataverse.
- ▶ **Challenge:** the outcome is severely **class-imbalanced** (onsets are rare), so accuracy is meaningless — evaluate with AUC / PR-AUC.

Extending with a random forest

Out-of-sample discrimination (area under the ROC curve)

Model	AUC
Logistic regression (original)	0.77
Random forest	0.91

- ▶ A large, well-documented jump (Muchlinski et al. 2016).
- ▶ The forest captures interactions and nonlinearities the additive logit cannot.
- ▶ A cautionary coda: later work warns against over-reading such gains — always inspect calibration and what drives the predictions.

Takeaway — rare events

- ▶ For rare, interactive outcomes, flexible learners can substantially out-discriminate a logit.
- ▶ But “higher AUC” is not the end of the story: check calibration, class-imbalance handling, and whether the lift is robust across folds/years.
- ▶ Pairs naturally with the Day-1 metric discussion (ROC vs. PR curves on imbalanced data).

Doing this yourself

A repeatable workflow

1. **Reproduce** the published regression from the replication archive (sanity check).
2. **Define the prediction task** and a held-out evaluation (k-fold CV).
3. **Fit a ladder of models:** OLS/GLM $\rightarrow$ regularized $\rightarrow$ trees/boosting.
4. **Compare out of sample** on a metric matched to the problem.
5. **Interpret** the best model (importance, partial dependence) and report honestly.

Common pitfalls

- ▶ Tuning on the test set (leakage) — always nest tuning inside the training folds.
- ▶ Comparing in-sample fit instead of out-of-sample prediction.
- ▶ Using accuracy on imbalanced outcomes.
- ▶ Forgetting to standardize before penalized or distance-based methods.
- ▶ Over-claiming: a predictive gain is not a causal claim.

Where to find more examples

- ▶ **Harvard Dataverse** and **OSF** host thousands of replication archives with data + code.
- ▶ Good candidates: a published *regression* on a substantive outcome, with enough features that nonlinearity/interactions are plausible.
- ▶ Reproduce first; extend second; let out-of-sample evidence decide.

Choosing among the penalized methods

If you want. . .	Use. . .
Stability with correlated predictors	Ridge (L2)
A sparse, interpretable model	Lasso (L1)
Sparsity <i>and</i> grouped correlated predictors	Elastic Net
The provably best small subset	Best subsets / ABESS
A nonparametric importance check	Random-forest importance

- Tune every penalty by cross-validation; never by in-sample fit.

Reproducibility checklist for an application

1. Pull data + code from the Dataverse/OSF archive; reproduce the published regression.
2. Fix a random seed; define the CV folds once and reuse them across models.
3. Standardize inside the training folds (not on the full data) for penalized/distance methods.
4. Report the metric that matches the problem (RMSE/ R^2 ; ROC-/PR-AUC).
5. Save the script so the comparison is itself reproducible.

Day 3 in one slide

- ▶ Regularization (ridge/lasso/elastic net) trades a little bias for much less variance.
- ▶ Lasso/ABESS additionally *select* variables — optimized for prediction, not inference.
- ▶ Whether ML beats the regression is an **empirical, out-of-sample** question: big for turnout, negligible for sparse-signal problems.
- ▶ Reproduce first, extend second, and let cross-validation decide.